

DIGITAL SAT — MATH

Theory Companion — Part 2

Advanced Math & Geometry / Trigonometry — Theory & Worked Problems

- 16 skills, one representative Hard/Very Hard worked problem per skill
- Advanced Math: Functions · Composite & Inverse · Transformations · Polynomials & Factoring · Poly Division & Rational Expressions · Quadratics · Radicals & Exponents · Exponential Functions · Radical & Rational Equations
- Geometry and Trigonometry: Angles & Triangles · Right Triangles & Trig · Radians & the Unit Circle · Circles · Volume & Surface Area
- Companion to the full 16-skill, 128-question Math Theory Study Companion — Greek terminology included for every skill

The 16 Skills

1. Functions — Notation, Domain, Range & Graphs
2. Composite & Inverse Functions
3. Function Transformations
4. Polynomial Operations & Special Products
5. Factoring Polynomials
6. Dividing Polynomials & Rational Expressions
7. Solving Quadratic Equations & the Discriminant
8. Quadratic Graphs, Roots & Word Problems
9. Radicals & Rational Exponents
10. Exponential Functions — Growth & Decay
11. Radical & Rational Equations
12. Angles, Parallel Lines & Triangle Fundamentals
13. Right Triangles & Trigonometry
14. Radians & the Unit Circle
15. Circles — Circumference, Arcs, Sectors, Equations & Theorems
16. Volume, Surface Area & Similarity of Solids



PART 1

Advanced Math

Functions & Notation · Composite & Inverse Functions · Transformations · Polynomial Operations & Factoring · Polynomial Division & Rational Expressions · Quadratic Equations, Graphs & Roots · Radicals, Rational Exponents & Exponential Functions · Radical & Rational Equations

Core Rules — In Detail

- Function Definition & Notation — $y = f(x)$ $f(x)$ is read 'f of x' — the output of function f when the input is x . Each input in the domain maps to exactly one output.
- Evaluating Functions — $f(a)$ = (substitute a for x) To evaluate f at a given input, substitute that input for the variable everywhere it appears, then simplify.
- The Vertical Line Test — A graph represents a function if and only if no vertical line intersects it more than once.
- Domain and Range — Domain = valid x -values. Range = resulting y -values. On a graph, the domain is read along the x -axis and the range along the y -axis.
- Domain Restrictions — Rational: denominator $\neq 0$. Even root: radicand ≥ 0 . Combine every restriction (multiple denominators, multiple roots) into one final domain statement.
- Piecewise Functions — $f(x) = \{\text{expr 1 if cond 1; expr 2 if cond 2; ...}\}$ Determine which condition the input satisfies, then use only the matching expression to evaluate.
- Rate of Change — Rate of change = $(f(x_2) - f(x_1)) / (x_2 - x_1)$ For a linear function this equals the slope; in context it tells you how much the output changes per one-unit increase in the input.
- Points on a Graph — Point (a, b) on $y = f(x)$ means $f(a) = b$. To find x such that $f(x) = k$, locate where the horizontal line $y = k$ crosses the graph — there may be more than one such x .

Watch Out: $f(a) = b$ means 'when the input is a , the output is b ' — students often flip which number is the input and which is the output when a word problem gives function notation. ⚠ For domain questions with more than one restriction (a root AND a denominator, for example), find each restriction separately and then combine them — don't stop after the first one.

Let $f(x) = 3x - k$ and $g(x) = x^2 + 2x$, where k is a positive integer. If $g(f(1)) = 24$, what is the value of $f(k)$?

Formula / Rule Used: Function composition: substitute $f(1)$ into g , then solve the resulting equation for the unknown constant.

Work through it, then advance for the full step-by-step solution.

Step 1: Compute $f(1)$ in terms of k : $f(1) = 3(1) - k = 3 - k$.

Step 2: Substitute this into g : $g(3 - k) = (3 - k)^2 + 2(3 - k) = 24$.

Step 3: Let $u = 3 - k$ to simplify: $u^2 + 2u - 24 = 0$.

Step 4: Factor the quadratic: $(u + 6)(u - 4) = 0$, so $u = -6$ or $u = 4$.

Step 5: Convert back to k using $k = 3 - u$: if $u = 4$, $k = -1$; if $u = -6$, $k = 9$.

Step 6: Since the problem states k is a positive integer, reject $k = -1$ and keep $k = 9$.

Step 7: So $f(x) = 3x - 9$. Compute $f(k) = f(9) = 3(9) - 9 = 27 - 9 = 18$.

Answer: $f(k) = 18$

Tip: When a composition gives a quadratic in disguise, solve for the inner variable first, then translate every root back to the original constant before applying any stated restriction (like 'positive integer') to throw out invalid roots.

Trap: A student who finds both roots of the quadratic but forgets to check the 'k is a positive integer' condition may keep $k = -1$, which produces a completely different (and wrong) final answer.

Tempting Wrong Answers — and Exactly What Went Wrong

$$f(k) = -2$$

Used the rejected root $u = 4$ (which gives $k = -1$) instead of checking the positive-integer condition, then computed $f(-1) = 3(-1) - (-1) = -2$ with that invalid k .

$$f(k) = 24$$

Stopped after finding $g(f(1)) = 24$ and reported that given value instead of continuing on to solve for k and then evaluate $f(k)$.

$$f(k) = 9$$

Correctly solved for $k = 9$ but then reported k itself as the final answer instead of substituting it back into $f(x) = 3x - 9$.

Shortcut: There's no shortcut around setting up and solving the quadratic in u , but working with $u = 3 - k$ instead of expanding everything in terms of k keeps the algebra one step shorter.

Desmos Tip: Graph $y = (3 - x)^2 + 2(3 - x) - 24$ (as a function of k) and find its zeros; the two x -intercepts are -1 and 9 , making it easy to see which one is the positive integer.

Core Rules — In Detail

- Composite Function Notation — $(f \circ g)(x) = f(g(x))$ This means: apply g first, then apply f to that result. Order matters — $f(g(x))$ is generally not the same as $g(f(x))$.
- Method 1 — Work Inside Out — Evaluate the inner function first at the given number, then plug that numeric result into the outer function.
- Method 2 — Find the Composite Expression First — Substitute the entire expression for $g(x)$ into every x in $f(x)$, simplify algebraically, and then evaluate at the desired value.
- Inverse Function Definition — $f(g(x)) = x$ and $g(f(x)) = x$ Two functions are inverses of each other if composing them (in either order) returns the original input. We write f^{-1} for the inverse of f .
- Finding an Inverse Algebraically — Steps: 1) $y = f(x)$ 2) swap x and y 3) solve for y 4) $y = f^{-1}(x)$ Swapping x and y reverses the roles of input and output, which is exactly what an inverse does.
- Inverse from a Table or Graph — $f^{-1}(b) = a$ if $f(a) = b$ To find f^{-1} of a value, look for that value in the output column (or y -coordinate) and read off the corresponding input.
- Graphical Relationship — The graphs of f and f^{-1} are reflections of each other across the line $y = x$.

Watch Out: In $f(g(x))$, always simplify the INNER function's output (or expression) completely before substituting into the outer function — don't substitute term-by-term. ⚠️ $(x + 2)^2 \neq x^2 + 4$ — a squared binomial substituted into a composite function must be expanded with FOIL/the perfect-square rule, never distributed directly.

A function f is defined implicitly by the equation $2f(x) - 3x = 8$. Let $g(x) = x + 5$. If f^{-1} denotes the inverse of f , find the value of $g(f^{-1}(7))$.

Formula / Rule Used: Solve an implicit equation for $f(x)$ first, find its inverse algebraically (swap x and y , solve for y), then evaluate the composite in the given order.

Work through it, then advance for the full step-by-step solution.

Step 1: Solve the implicit equation for $f(x)$: $2f(x) - 3x = 8$, so $2f(x) = 3x + 8$, and $f(x) = (3x + 8)/2$.

Step 2: Find the inverse of f by replacing $f(x)$ with y : $y = (3x + 8)/2$.

Step 3: Swap x and y : $x = (3y + 8)/2$.

Step 4: Solve for y : $2x = 3y + 8$, so $3y = 2x - 8$, so $y = (2x - 8)/3$. This means $f^{-1}(x) = (2x - 8)/3$.

Step 5: Evaluate $f^{-1}(7)$: $f^{-1}(7) = (2(7) - 8)/3 = (14 - 8)/3 = 6/3 = 2$.

Step 6: Evaluate g at this result: $g(f^{-1}(7)) = g(2) = 2 + 5 = 7$.

Answer: $g(f^{-1}(7)) = 7$

Tip: When a function is given implicitly, isolate $f(x)$ completely first before doing anything else — you cannot find an inverse or compose functions from an equation that isn't solved for $f(x)$ yet.

Trap: The notation $f^{-1}(7)$ invites two classic errors here: treating it as $1/f(7)$ (reciprocal) or evaluating the functions in the wrong order by misreading which one is 'inside'.

Tempting Wrong Answers — and Exactly What Went Wrong

16/3 Reversed the composition order, computing $f^{-1}(g(7))$ instead of $g(f^{-1}(7))$ — first found $g(7) = 12$, then applied f^{-1} to that.

37/3 Made a sign error isolating y when finding the inverse, solving $x = (3y+8)/2$ as $y = (2x+8)/3$ instead of $y = (2x-8)/3$, then continued the rest of the problem with that wrong inverse.

39/2 Skipped finding the inverse entirely and evaluated $f(7)$ instead of $f^{-1}(7)$, using $f(7) = (3(7) + 8)/2 = 2\frac{1}{2}$ as the input to g .

Shortcut: Since g is just 'add 5', once you know $f^{-1}(7) = 2$ the final answer is simply $2 + 5$ — no need to write a separate composition line.

Desmos Tip: Graph $y = (3x + 8)/2$ and its reflection $y = (2x - 8)/3$ across $y=x$ to confirm they're inverses, then trace the second graph to $x=7$ to read off $f^{-1}(7) = 2$.

Core Rules — In Detail

- Vertical Shift — $y = f(x) + k$ Shifts the graph k units — up if $k > 0$, down if $k < 0$. Every y -value changes by k ; x -values are untouched.
- Horizontal Shift — $y = f(x + h)$ Shifts the graph h units LEFT if $h > 0$, and RIGHT if $h < 0$ — the opposite direction you'd expect from the sign.
- Reflections — $y = -f(x)$ reflects across the x -axis. $y = f(-x)$ reflects across the y -axis. Negating the whole output flips the graph vertically; negating the input flips it horizontally.
- Vertical Stretch/Compression — $y = a \cdot f(x)$ If $|a| > 1$, the graph stretches vertically (taller). If $0 < |a| < 1$, it compresses (shorter). If $a < 0$, it also reflects across the x -axis.
- Combined Transformation — $y = a \cdot f(x - h) + k$ Read transformations from the inside out: horizontal shift by h , vertical stretch/compression and reflection by a , then vertical shift by k .

Watch Out: Horizontal shifts are backwards from intuition: $f(x+2)$ shifts LEFT 2, not right — to keep the same output as the original at $x=0$, the new input must be 2 less. ⚠ When a problem gives you a transformed graph (like $y = f(x+2)+1$) and asks about the ORIGINAL $f(x)$, work backward: undo the transformations in reverse order.

A function f is defined only by the table below, for integer inputs from -4 to 5:

x : -4 -3 -2 -1 0 1 2 3 4 5

$f(x)$: 8 -3 0 1 2 4 9 1 6 -2

Let $g(x) = a \cdot f(x + 3) - 5$ for some constant a . If $g(1) = 7$, what is $g(-4)$?

Formula / Rule Used: Combined transformation $g(x) = a \cdot f(x+h)+k$: horizontal shift left by h ($h>0$), vertical stretch by a , vertical shift by k .

Work through it, then advance for the full step-by-step solution.

Step 1: Substitute $x = 1$ into $g(x) = a \cdot f(x+3) - 5$: $g(1) = a \cdot f(1+3) - 5 = a \cdot f(4) - 5$.

Step 2: From the table, $f(4) = 6$, so $g(1) = 6a - 5$.

Step 3: Set this equal to the given value: $6a - 5 = 7$.

Step 4: Solve for a : $6a = 12$, so $a = 2$.

Step 5: Now find $g(-4)$ using the same rule: $g(-4) = a \cdot f(-4+3) - 5 = 2 \cdot f(-1) - 5$.

Step 6: From the table, $f(-1) = 1$.

Step 7: Compute: $g(-4) = 2(1) - 5 = 2 - 5 = -3$.

Answer: $g(-4) = -3$

Tip: When a transformation formula has an unknown constant, first solve for that constant using the one given point, then re-apply the FULL rule (including the input shift) to the new input — don't skip straight to plugging the new x into f .

Trap: It's tempting to plug $x = -4$ directly into f from the table once a is known, forgetting that g still shifts the input by $+3$ first, which silently swaps $f(-1)$ for $f(-4)$ and wrecks the final answer.

Tempting Wrong Answers — and Exactly What Went Wrong

19

Treated $g(x) = a \cdot f(x) - 5$ without applying the +3 input shift at all, using $f(1)=4$ and $f(-4)=8$ instead of $f(4)$ and $f(-1)$, so a solved to 3 and the final value came out to $3(8)-5=19$.

7

Correctly found $a = 2$, but added 5 instead of subtracting it in the last step ($2(1)+5=7$), mixing up the sign of the vertical shift.

11

Correctly solved $a = 2$, but then forgot to re-apply the +3 shift when evaluating $g(-4)$, plugging $f(-4)=8$ directly instead of $f(-4+3)=f(-1)=1$, giving $2(8)-5=11$.

Shortcut: Once $a = 2$ is found, skip rewriting the whole equation again — just compute the shifted input ($-4+3 = -1$), read $f(-1)$ from the table, and plug it straight into $2(\cdot) - 5$.

Desmos Tip: Enter the table's x -values and $f(x)$ -values as two lists in Desmos, verify $a=2$ makes point $(1,7)$ satisfy g , then read g at the shifted input -1 to confirm -3 .

Core Rules — In Detail

- Polynomial Vocabulary — $a_nx^n + \dots + a_1x + a_0$ Degree = highest exponent; leading coefficient = coefficient of the highest-degree term; like terms share the same variable AND exponent.
- Adding/Subtracting Polynomials — Combine only like terms, adding or subtracting their coefficients. When subtracting, distribute the negative sign to EVERY term of the second polynomial first.
- Distributive Property / FOIL — $a(b+c) = ab + ac$. $(a+b)(c+d) = ac+ad+bc+bd$ For binomials, FOIL (First, Outer, Inner, Last) is the distributive property applied twice; for larger polynomials, multiply every term in the first by every term in the second.
- Square of a Sum — $(a + b)^2 = a^2 + 2ab + b^2$ Square the first term, double the product of the two terms, square the last term.
- Square of a Difference — $(a - b)^2 = a^2 - 2ab + b^2$ Same pattern as the sum, but the middle term is negative.
- Difference of Squares — $(a + b)(a - b) = a^2 - b^2$ The middle terms cancel exactly — there is never a middle term in the expanded result.

Watch Out: $(x + 5)^2$ is NEVER $x^2 + 25$ — the middle term $2ab$ is real and is the single most common polynomial-operations mistake on the SAT. 
When subtracting one polynomial from another, every single term in the second polynomial changes sign — forgetting even one term is the most common subtraction error.



If $(x + p)^2 = x^2 + 26x + q$ for all real numbers x , where p and q are positive constants, what is the value of $q - p$?

Formula / Rule Used: Perfect square expansion: $(a + b)^2 = a^2 + 2ab + b^2$, matched coefficient by coefficient.

Work through it, then advance for the full step-by-step solution.

Step 1: Expand $(x + p)^2$ using the special product rule: $(x + p)^2 = x^2 + 2px + p^2$.

Step 2: Since this equals $x^2 + 26x + q$ for every x , the x -coefficients must match: $2p = 26$.

Step 3: Solve for p : $p = 13$.

Step 4: Match the constant terms: $q = p^2$.

Step 5: Substitute $p = 13$: $q = 13^2 = 169$.

Step 6: Compute $q - p = 169 - 13 = 156$.

Answer: $q - p = 156$

Tip: When a perfect-square expansion is set equal to a general trinomial, match the x -coefficient first to find the base constant, then square it to get the last term — don't try to read q off directly.

Trap: It's tempting to treat q as if it appears directly in the expansion (like $q = 26$, copying the x -coefficient) instead of recognizing that q is actually p^2 , which requires an extra squaring step after finding p .



Tempting Wrong Answers — and Exactly What Went Wrong

$q - p = 650$

Set $2p = 26$ but forgot to divide by 2, using $p = 26$ instead of 13, then computed $q = 26^2 = 676$, giving $q - p = 650$.

$q - p = -156$

Correctly found $p = 13$ and $q = 169$, but subtracted in the wrong order ($p - q$ instead of $q - p$), getting -156.

$q - p = 13$

Mistakenly used $q = 2p$ (mixing up the linear and squared terms of the identity) instead of $q = p^2$, getting $q = 26$ and $q - p = 26 - 13 = 13$.

Shortcut: Once $2p = 26$ gives $p = 13$, there's no need to re-expand anything symbolically — just square 13 directly to get $q = 169$.

Desmos Tip: Graph $y = (x + 13)^2$ and $y = x^2 + 26x + 169$ — the two curves should perfectly overlap, confirming $p = 13$ and $q = 169$ before computing the difference.

Core Rules — In Detail

- Factor Out the GCF First — Always check for a greatest common factor (of both the numbers and the variables — using the SMALLEST exponent of each variable present) before trying any other method.
- Factoring $x^2 + bx + c$ (leading coefficient 1) — Find r, s with $r \times s = c$ and $r + s = b$, then $(x+r)(x+s)$ List factor pairs of c and check which pair sums to b — track signs carefully when c is negative.
- The AC Method for $ax^2 + bx + c$ ($a \neq 1$) — Find factors of $a \times c$ that sum to b , rewrite bx as their sum, then factor by grouping. This turns a hard trinomial into two easier grouping steps.
- Difference of Squares — $a^2 - b^2 = (a - b)(a + b)$ Recognizable whenever two perfect squares are subtracted — check twice, since $a^4 - 81$ factors as a difference of squares AND then factors again.
- Perfect Square Trinomial — $a^2 \pm 2ab + b^2 = (a \pm b)^2$ Check: is the first term a perfect square, the last term a perfect square, and the middle term exactly twice their product?
- Factoring by Grouping (Cubics) — Group the first two and last two terms, factor each group separately, then factor out the common binomial that remains.
- Multiplicity of Roots — The exponent on a factor is that root's multiplicity. Odd multiplicity: the graph crosses the x-axis. Even multiplicity: the graph touches the x-axis and turns back.

Watch Out: A polynomial isn't 'fully factored' just because you found one difference of squares — always check whether either resulting factor can be factored again. ⚠️ $x^2 + 9$ (a sum of squares) does NOT factor over the real numbers — don't force a factorization that doesn't exist.

The equation $12x^2 - 17x = 40$ has two real solutions. What is the positive difference between them?

Formula / Rule Used: Rearrange to standard form $ax^2 + bx + c = 0$, then factor by the AC method (grouping) when a is not 1.

Work through it, then advance for the full step-by-step solution.

Step 1: Rearrange to standard form: $12x^2 - 17x - 40 = 0$.

Step 2: For the AC method, multiply a and c: $12 \times (-40) = -480$.

Step 3: Find two numbers that multiply to -480 and add to -17: -32 and 15 work, since $(-32)(15) = -480$ and $-32 + 15 = -17$.

Step 4: Rewrite the middle term and group: $12x^2 - 32x + 15x - 40 = 0$.

Step 5: Factor each pair: $4x(3x - 8) + 5(3x - 8) = 0$.

Step 6: Factor out the common binomial: $(4x + 5)(3x - 8) = 0$.

Step 7: Solve each factor: $4x + 5 = 0$ gives $x = -5/4$; $3x - 8 = 0$ gives $x = 8/3$.

Step 8: Find the positive difference: $8/3 - (-5/4) = 32/12 + 15/12 = 47/12$.

Answer: 47/12

Tip: Before applying the AC method, always move every term to one side so the equation reads $ax^2 + bx + c = 0$ -- factoring a rearranged equation directly produces the wrong factors.

Trap: A student who factors $12x^2 - 17x = 40$ without rearranging might factor the left side alone as $x(12x - 17)$ and effectively solve $12x^2 - 17x = 0$, 'losing' the 40 and getting the false solutions $x=0$ and $x = \frac{17}{12}$.

Tempting Wrong Answers — and Exactly What Went Wrong

17/12

Forgot to move the 40 to the left side and instead factored $12x^2 - 17x = x(12x - 17) = 0$, getting the incorrect solutions $x = 0$ and $x = \frac{17}{12}$.

17/3

Chose the wrong number pair in the AC method (8 and -60, which multiply to -480 but add to -52, not -17), leading to incorrect binomials $(4x-20)(3x+2)$ and roots $x=5$ and $x=-2/3$.

13/7

Found the correct roots $8/3$ and $-5/4$ but added the fractions incorrectly by combining numerators and denominators separately $(8-(-5))/(3+4)$ instead of finding a common denominator.

Shortcut: Skip full factoring for the difference: $|r_1 - r_2| = \sqrt{(b^2 - 4ac)/a} = \sqrt{((-17)^2 - 4(12)(-40))/12} = \sqrt{2209/12} = \frac{47}{12}$, directly.

Desmos Tip: Graph $y = 12x^2 - 17x - 40$ and use the calculator's intersection/root feature to find both x-intercepts, then subtract them directly.

Core Rules — In Detail

- Dividing by a Monomial — $(a+b+c)/d = a/d + b/d + c/d$ Split into separate fractions and simplify each term individually.
- Polynomial Long Division — Divide leading terms to get the next quotient term, multiply that term by the whole divisor, subtract, bring down the next term, and repeat until the remainder's degree is less than the divisor's.
- Simplifying Rational Expressions — Factor the numerator and denominator completely, then cancel any common factors — the simplified expression excludes whatever values made the ORIGINAL denominator zero.
- Multiplying/Dividing Rational Expressions — Factor everything first, then cancel common factors across the whole product. For division, multiply by the reciprocal of the second fraction.
- Adding/Subtracting Rational Expressions — Find the least common denominator (LCD), rewrite each fraction over it, then combine the numerators.
- Solving Rational Equations — Multiply every term by the LCD to clear all denominators, solve the resulting equation, then discard any solution that makes an original denominator zero (extraneous).
- Complex Fractions — Either combine the numerator and denominator into single fractions and then divide, or multiply the entire complex fraction by the LCD of all the inner fractions.

Watch Out: Canceling a factor changes the domain unless you track it — the simplified expression is only equivalent to the original for x-values that didn't make the original denominator zero. ⚠ Every rational-equation solution must be checked against every original denominator, not just the one that produced it — an extraneous solution can come from any denominator in the problem.

When the polynomial $3x^3 + ax^2 - 7x + 4$ is divided by $(x - 2)$, the quotient is $3x^2 + 5x + 3$ and the remainder is 10. Find the value of a .

Formula / Rule Used: Division Algorithm: Dividend = Divisor x Quotient + Remainder; equivalently, Remainder Theorem: dividing by $(x-k)$ leaves remainder $P(k)$.

Work through it, then advance for the full step-by-step solution.

Step 1: Recall the division algorithm: Dividend = Divisor x Quotient + Remainder.

Step 2: Recognize the faster path: dividing by $(x-2)$ means the Remainder Theorem applies, so $P(2)$ equals the remainder, 10.

Step 3: Substitute $x=2$ into $3x^3 + ax^2 - 7x + 4$: $3(2)^3 + a(2)^2 - 7(2) + 4 = 3(8) + 4a - 14 + 4$.

Step 4: Simplify the constants: $24 - 14 + 4 = 14$, so $P(2) = 14 + 4a$.

Step 5: Set this equal to the given remainder: $14 + 4a = 10$.

Step 6: Solve: $4a = -4$, so $a = -1$.

Step 7: Check by expansion: $(x - 2)(3x^2 + 5x + 3) + 10 = 3x^3 - x^2 - 7x + 4$, which matches $3x^3 + ax^2 - 7x + 4$ exactly when $a=-1$.

Answer: $a = -1$

Tip: When a problem gives you a quotient AND a remainder and asks for a missing coefficient in the dividend, plug the divisor's root into the dividend (Remainder Theorem) instead of expanding divisor times quotient by hand.

Trap: Students who don't trust the Remainder Theorem often expand $(x - 2)(3x^2 + 5x + 3)$ term by term, and a sign slip in that expansion (especially with the -2 times 5x term) is far more likely than the one-line substitution.

Tempting Wrong Answers — and Exactly What Went Wrong

a = -8

Computed $-7(2)$ as $+14$ instead of -14 (dropped the negative sign), giving $24+4a+14+4=10$ and solving $4a=-32$.

a = -3.5

Confused the Remainder Theorem with the Factor Theorem and set $P(2)$ equal to 0 (as if the divisor left no remainder) instead of equal to the given remainder 10.

a = 0.5

Miscomputed 2^3 as $2 \times 3 = 6$ instead of $2 \times 2 \times 2 = 8$, corrupting the first term of $P(2)$ to 18 instead of 24.

Shortcut: The Remainder Theorem already IS the shortcut here — there's no need to ever multiply out $(x - 2)(3x^2 + 5x + 3)$; a one-line substitution of $x=2$ gets the answer directly.

Desmos Tip: Graph $y = 3x^3 - x^2 - 7x + 4$ (using $a=-1$) and check that the y -value at $x=2$ reads exactly 10, matching the given remainder.

Core Rules — In Detail

- Solving by Factoring — $ax^2 + bx + c = 0 \rightarrow \text{factor} \rightarrow \text{set each factor to 0}$ Fastest method whenever the trinomial factors nicely; always check that the equation is set equal to zero first.
- Solving by Taking Square Roots — If $x^2 = k$, then $x = \pm\sqrt{k}$. Use when the equation is (or can be isolated to) a squared term equal to a constant — never forget the $\pm$ sign.
- Solving by Completing the Square — Move the constant to the other side, take half the x-coefficient, square it, add that to both sides, then factor the left side as a perfect square.
- The Quadratic Formula — $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ Works for ANY quadratic equation, especially when factoring is difficult or impossible — always identify a, b, and c carefully first.
- The Discriminant — $D = b^2 - 4ac$ $D > 0$: two distinct real solutions. $D = 0$: exactly one real solution (a repeated root). $D < 0$: no real solutions.

Watch Out: Forgetting the $\pm$ when taking a square root loses half of the valid solutions — this is one of the most common quadratic-solving errors.

! Before using the discriminant or quadratic formula, the equation MUST be in standard form $ax^2+bx+c=0$ — rearrange first if it isn't.

The equation $3x^2 - kx + 12 = 0$, where k is a positive constant, has a discriminant equal to 25. What is the value of k ?

Formula / Rule Used: Discriminant $D = b^2 - 4ac$.

Work through it, then advance for the full step-by-step solution.

Step 1: Identify $a = 3$, $b = -k$, and $c = 12$ in the equation $3x^2 - kx + 12 = 0$.

Step 2: Write the discriminant symbolically: $D = b^2 - 4ac = (-k)^2 - 4(3)(12)$.

Step 3: Simplify: $D = k^2 - 144$.

Step 4: Set this equal to the given discriminant value: $k^2 - 144 = 25$.

Step 5: Solve for k^2 : $k^2 = 169$.

Step 6: Take the square root of both sides, remembering both signs: $k = \pm 13$.

Step 7: Since the problem states k is positive, discard $k = -13$ and keep $k = 13$.

Answer: $k = 13$

Tip: Write the discriminant expression symbolically first (in terms of the unknown), then set it equal to the target value and solve like any other equation.

Trap: A rushed student who solves $k^2 = 169$ might report both roots or grab the negative one without noticing the problem restricts k to positive values, since -13 is a perfectly valid algebraic solution to $k^2 = 169$.

Tempting Wrong Answers — and Exactly What Went Wrong

k = -13

Solved $k^2 = 169$ correctly but ignored the problem's stated restriction that k is positive, reporting the negative root instead.

No real value of k works

Used $D = b^2 + 4ac$ instead of $D = b^2 - 4ac$ (sign error), getting $k^2 + 144 = 25$, i.e. $k^2 = -119$, which incorrectly appears to have no real solution.

k = 169

Forgot to square $b = -k$ when writing the discriminant, using $D = k - 4ac$ instead of $D = k^2 - 4ac$, so $k - 144 = 25$ gave $k = 169$.

Shortcut: Since only the positive value of k is requested, you can skip writing ' $\pm$ ' altogether and just take the positive square root of 169 once you reach $k^2 = 169$.

Desmos Tip: Graph $y = 3x^2 - 13x + 12$ on Desmos and confirm it crosses the x-axis at two distinct points, consistent with a positive discriminant.

Core Rules — In Detail

- Standard Form — $y = ax^2 + bx + c$ c is the y-intercept; the vertex's x-coordinate is $x_v = -b/2a$.
- Factored Form — $y = a(x - p)(x - q)$ p and q are the x-intercepts (roots); the vertex's x-coordinate is the average of the roots, $(p+q)/2$.
- Vertex Form — $y = a(x - h)^2 + k$ (h, k) is the vertex directly — the minimum point if $a > 0$, the maximum point if $a < 0$.
- Completing the Square to Reach Vertex Form — Factor a out of the first two terms, add and subtract $(\text{half the new x-coefficient})^2$ inside, then factor the resulting perfect square.
- Sum and Product of Roots (Vieta's Formulas) — Sum = $-b/a$. Product = c/a . Lets you find a combination of the roots without ever solving the equation directly.
- Quadratic Word Problems — Translate the scenario into a quadratic equation (area = length×width, height = $-16t^2 + v_0t + h_0$, etc.), solve, and discard any solution that isn't physically valid (negative length, negative time).

Watch Out: A parabola opening downward ($a < 0$) has a MAXIMUM at its vertex, not a minimum — always check the sign of a before describing the vertex. ⚠ In a word problem, always discard algebraically valid solutions that make no real-world sense (negative width, negative time before an event starts).

A quadratic function has vertex $(3, -16)$. Its graph crosses the x -axis at two points, and the product of the two x -intercepts is -7 . If the function is written as $y = a(x - 3)^2 - 16$, what is the value of a ?

Formula / Rule Used: Vieta's formulas: sum of roots $= -B/A$, product of roots $= C/A$, applied after expanding vertex form to standard form.

Work through it, then advance for the full step-by-step solution.

Step 1: Expand the vertex form: $y = a(x - 3)^2 - 16 = a(x^2 - 6x + 9) - 16 = ax^2 - 6ax + (9a - 16)$.

Step 2: Identify the standard-form coefficients: $A = a$, $B = -6a$, $C = 9a - 16$.

Step 3: Apply Vieta's product-of-roots formula: $\text{product} = C/A = (9a-16)/a$.

Step 4: Set this equal to the given product, -7: $(9a-16)/a = -7$.

Step 5: Multiply both sides by a : $9a - 16 = -7a$.

Step 6: Solve: $9a + 7a = 16$, so $16a = 16$, and $a = 1$.

Answer: $a = 1$

Tip: Whenever a parabola is given in vertex form but the question refers to its roots (sum or product), expand to standard form first so you can apply Vieta's formulas directly on A, B, and C.

Trap: It's tempting to treat the product of roots as $-C/A$ (mirroring the sum formula's minus sign) instead of the correct C/A , which produces $a = 8$ instead of $a = 1$.

Tempting Wrong Answers — and Exactly What Went Wrong

8

Used $-C/A$ instead of C/A for the product of roots (mixing up the sign pattern from the sum formula), solving $-(9a-16)/a = -7$ to get $a = 8$.

-1

Made a sign error moving terms, writing $9a + 16 = -7a$ instead of $9a - 16 = -7a$, which gives $16a = -16$ and $a = -1$.

-8

Expanded $(x - 3)^2$ incorrectly as $x^2 - 9$ (misapplying a difference-of-squares pattern instead of squaring the binomial), which changed C and produced $a = -8$ instead of $a = 1$.

Shortcut: Since the vertex's x -coordinate already tells you the roots' sum is $2(3) = 6$, you can skip straight to setting up the product $= C/A$ equation without separately re-deriving or checking the sum.

Desmos Tip: Graph $y = (x - 3)^2 - 16$ (using $a = 1$) and confirm the x -intercepts are at $x = -1$ and $x = 7$, whose product $(-1)(7) = -7$ matches the given condition.

Core Rules — In Detail

- Radical-Exponent Conversion — $\sqrt[n]{a^m} = a^{(m/n)}$ The denominator of the fractional exponent is the root (index); the numerator is the power.
- Properties of Radicals — $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ $\sqrt[n]{a/b} = \sqrt[n]{a} / \sqrt[n]{b}$ Valid for non-negative radicands when the index is even; valid for all real radicands when the index is odd.
- Simplifying Radicals — Factor out the largest perfect power (square, cube, etc.) hiding inside the radicand and pull it out.
- Adding/Subtracting Radicals — Only radicals with the SAME index and SAME radicand can be combined, exactly like combining like terms — simplify each radical first, since disguised matches are common.
- Rationalizing Denominators — Single term: multiply by the radical. Binomial: multiply by the conjugate. The conjugate of $a+\sqrt{b}$ is $a-\sqrt{b}$; multiplying a binomial radical denominator by its conjugate eliminates the radical via the difference-of-squares pattern.
- Exponent Rules with Rational Exponents — $a^{(m/n)} \cdot a^{(p/n)} = a^{((m+p)/n)}$ The product, quotient, and power rules for integer exponents apply identically to fractional exponents.

Watch Out: An even-index root of a negative radicand is not a real number — $\sqrt[4]{-4}$ has no real value, but cube roots of negative numbers ARE real: $\sqrt[3]{-8} = -2$. ⚠ Before adding or subtracting radicals, simplify each one first — $\sqrt{45} + \sqrt{20}$ look unrelated until simplified to $3\sqrt{5} + 2\sqrt{5}$.



Simplify the expression $[(a^{2/3} \cdot b^{-1/2})^3] / (a^{1/2} \cdot b^{-1})$, then evaluate the result for $a = 4$ and $b = 9$.

Formula / Rule Used: Rational exponent rules: $(x^m)^n = x^{m \cdot n}$; $x^m / x^n = x^{m - n}$.

Work through it, then advance for the full step-by-step solution.

Step 1: Apply the power rule to the numerator: $(a^{(\frac{2}{3})})^3 = a^{(\frac{2}{3} \cdot 3)} = a^2$, and $(b^{(-\frac{1}{2})})^3 = b^{(-\frac{3}{2})}$. The numerator becomes $a^2 \cdot b^{(-\frac{3}{2})}$.

Step 2: Divide by the denominator's a-term using the quotient rule: $a^2 / a^{(\frac{1}{2})} = a^{(2 - \frac{1}{2})} = a^{(\frac{3}{2})}$.

Step 3: Divide by the denominator's b-term: $b^{(-\frac{3}{2})} / b^{(-1)} = b^{(-\frac{3}{2} - (-1))} = b^{(-\frac{1}{2})}$.

Step 4: The fully simplified expression is $a^{(\frac{3}{2})} \cdot b^{(-\frac{1}{2})}$, which is the same as $a^{(\frac{3}{2})} / b^{(\frac{1}{2})}$.

Step 5: Evaluate $a^{(\frac{3}{2})}$ at $a = 4$: $a^{(\frac{3}{2})} = (a^{(\frac{1}{2})})^3 = (\sqrt{4})^3 = 2^3 = 8$.

Step 6: Evaluate $b^{(\frac{1}{2})}$ at $b = 9$: $\sqrt{9} = 3$.

Step 7: Combine the results: $8/3$.

Answer: $8/3$

Tip: Fully simplify the exponents symbolically first — combine everything into a single power of a and a single power of b — before substituting numbers; plugging in early forces you to compute ugly fractional powers like $4^{(\frac{3}{2})}$ instead of clean ones like $4^{(\frac{3}{2})}$.

Trap: A student who substitutes $a = 4$ and $b = 9$ immediately, before simplifying the exponent algebra, ends up needing to compute things like $4^{(\frac{3}{2})}$ and $9^{(-\frac{1}{2})}$ directly inside a cubed expression and often mishandles the negative exponent on b, landing on $\frac{3}{8}$ instead of $\frac{8}{3}$.

Tempting Wrong Answers — and Exactly What Went Wrong

3/8

Simplified the exponents correctly but mishandled the negative sign on $b^{(-\frac{1}{2})}$, inverting the final fraction so b ended up in the numerator instead of the denominator.

24

Correctly got $a^{\frac{2}{3}} = 8$ but treated $b^{(-\frac{1}{2})}$ as $b^{\frac{1}{2}}$ (dropped the negative sign entirely), computing $8 * 3 = 24$ instead of $\frac{8}{3}$.

8/9

Correctly simplified a's exponent to $a^{\frac{2}{3}} = 8$ but never finished subtracting the b exponents down to $-\frac{1}{2}$, plugging in $b = 9$ as if its exponent were still -1, giving $\frac{8}{9}$.

Shortcut: Track the exponent on a and the exponent on b as two separate running totals (starting at $\frac{2}{3} * 3 = 2$ and $-\frac{1}{2} * 3 = -\frac{3}{2}$) and subtract the denominator's exponents from each as you go, instead of writing out the full expanded expression with both variables tangled together.

Desmos Tip: Enter $(a^{\frac{2}{3}} \cdot b^{(-\frac{1}{2})})^3 / (a^{\frac{1}{2}} \cdot b^{(-1)})$ into Desmos with $a = 4$ and $b = 9$ typed in directly to numerically confirm the result is $\frac{8}{3}$ (about 2.667).

Core Rules — In Detail

- General Exponential Form — $y = a \cdot b^x$ a is the initial value (value at $x=0$); b is the growth/decay factor. If $b>1$, growth; if $0<b<1$, decay.
- Percent Change Connection — Growth: $A = P(1+r)^t$. Decay: $A = P(1-r)^t$. r is the percent rate as a decimal; the base b in the general form equals $(1+r)$ for growth or $(1-r)$ for decay.
- Key Features of the Basic Graph — y-intercept $(0, a)$; horizontal asymptote $y = 0$ Domain is all real numbers; range is $y > 0$ for a positive initial value — the graph never touches its asymptote.
- Solving with the Same-Base Method — If $a^m = a^n$ (same base a), then $m = n$. Rewrite both sides with the same base when possible (e.g. 9 and 27 both as powers of 3), then set the exponents equal and solve.
- Transformations of Exponential Graphs — $y = b^x + k$ shifts the asymptote to $y = k$. $y = b^{(x-h)}$ shifts horizontally. A vertical shift changes the horizontal asymptote itself, not just the graph's position.

Watch Out: A vertical shift on an exponential graph moves the asymptote too — $y = 2^x + 3$ has asymptote $y = 3$, not $y = 0$. ⚠ When rewriting exponential equations to a common base, remember that raising a power to a power MULTIPLIES exponents — a very common slip when converting bases like $9 = 3^2$.

A tech startup based in Thessaloniki has revenue that grows exponentially over time. Company records show that revenue was €120,000 in year 2 after launch and €172,800 in year 4 after launch. The company currently employs 45 people. Assuming the same exponential growth pattern continues, what will the company's revenue be in year 7 after launch, to the nearest euro?

Formula / Rule Used: Exponential growth model $A = a \cdot b^t$; divide two known outputs to isolate a power of b .

Work through it, then advance for the full step-by-step solution.



Step 1: Let $R(t) = a \cdot b^t$ model revenue t years after launch, where a is the initial revenue and b is the annual growth factor.

Step 2: Divide the year-4 revenue by the year-2 revenue to eliminate a : $(a \cdot b^4) / (a \cdot b^2) = b^2 = 172,800 / 120,000 = 1.44$.

Step 3: Solve for b : $b = \sqrt{1.44} = 1.2$, so revenue grows by 20% each year.

Step 4: Rather than solving for a , project forward directly from year 4: since $7 - 4 = 3$ years pass, $R(7) = R(4) \cdot b^3$.

Step 5: Compute b^3 : $1.2^3 = 1.2 \cdot 1.2 \cdot 1.2 = 1.728$.

Step 6: Multiply: $R(7) = 172,800 \times 1.728 = 298,598.4$.

Step 7: Round to the nearest euro: $R(7) \approx \text{€}298,598$. (The fact that the company has 45 employees plays no role in the calculation.)

Answer: €298,598

Tip: When two outputs of an exponential model are given at two different input values, divide them to isolate a power of b — never average or subtract the two values.

Trap: A rushed student takes 1.44 itself as the annual growth factor instead of recognizing it as b^2 and taking its square root, which compounds the growth far too fast over the remaining years.

Tempting Wrong Answers — and Exactly What Went Wrong

≈€743,046

Used 1.44 directly as the one-year growth factor instead of taking its square root to isolate the true annual multiplier $b = 1.2$, so every remaining year compounded too aggressively.

€252,000

Modeled the growth as linear, adding a flat $(172,800 - 120,000) / 2 = 26,400$ per year, instead of recognizing that revenue changes multiplicatively, not additively.

€358,318

Correctly found $b = 1.2$ and used $R(4)$ as the base, but miscounted the gap from year 4 to year 7 as 4 years instead of 3, computing $R(4) \cdot 1.2^4$ instead of $R(4) \cdot 1.2^3$.

Shortcut: Once b is found, project forward from the later known data point ($R(4)$) using b raised to the remaining number of years, instead of solving for the initial value a first and restarting from year 0.

Desmos Tip: Plot the points (2, 120000) and (4, 172800), fit $y = a \cdot b^x$ by adjusting a and b until the curve passes through both, then trace to $x = 7$ to read the projected revenue.

Core Rules — In Detail

- Solving Radical Equations — Isolate the radical on one side, raise both sides to the power matching the index, solve the resulting equation, then plug every solution back into the ORIGINAL equation.
- Extraneous Solutions from Radicals — Raising both sides to an EVEN power can introduce solutions that don't satisfy the original equation — always check. Odd-index roots (cube roots) never create this problem.
- Radical Equations with Two Radicals — Isolate one radical, square, then isolate the remaining radical and square again if needed.
- Solving Rational Equations — Multiply every term by the least common denominator (LCD) to clear all fractions at once, solve the resulting polynomial equation, then discard any solution making an original denominator zero.
- Complex Fractions — Multiply the entire complex fraction (top and bottom) by the LCD of every inner fraction to clear all the small fractions in one step.

Watch Out: 'Solve, then check' is not optional for radical equations raised to an even power — a fully correct algebraic solution can still be extraneous. ⚠ In a rational equation, a value is extraneous if it zeroes out ANY original denominator — check every denominator in the problem, not just the most obvious one.

Find all real values of x that satisfy: $\sqrt{2x + 9} - \sqrt{x + 4} = 1$.

Formula / Rule Used: For equations with two radicals, isolate one radical, square, isolate the remaining radical, then square again.

Work through it, then advance for the full step-by-step solution.

Step 1: Isolate one radical by moving the other to the right side: $\sqrt{2x + 9} = 1 + \sqrt{x + 4}$.

Step 2: Square both sides: $2x + 9 = 1 + 2\sqrt{x + 4} + (x + 4) = x + 5 + 2\sqrt{x + 4}$.

Step 3: Isolate the remaining radical: $2x + 9 - x - 5 = 2\sqrt{x + 4}$, so $x + 4 = 2\sqrt{x + 4}$.

Step 4: Square both sides again: $(x + 4)^2 = 4(x + 4)$.

Step 5: Move everything to one side and factor out $(x + 4)$: $(x + 4)^2 - 4(x + 4) = (x + 4)$.

Step 6: So $x = -4$ or $x = 0$.

Step 7: Check $x = -4$ in the original equation: $\sqrt{2(-4) + 9} - \sqrt{-4 + 4} = \sqrt{1} - \sqrt{0} = 1 - 0 = 1$. Valid.

Step 8: Check $x = 0$ in the original equation: $\sqrt{9} - \sqrt{4} = 3 - 2 = 1$. Also valid.

Step 9: Both candidates check out, so both are genuine solutions.

Answer: $x = -4$ or $x = 0$

Tip: With two radicals, isolate ONE of them before the first squaring — never square while both radicals sit on the same side, or the cross-term won't cancel cleanly.

Trap: A student who squares the original equation as $(\sqrt{2x + 9} - \sqrt{x + 4})^2 = 1$ without isolating first gets a messier cross term involving $\sqrt{((2x + 9)(x + 4))}$; moving pieces around from there is where an extra sign flip usually sneaks in.

Tempting Wrong Answers — and Exactly What Went Wrong

$x = 0$

Found both roots from the second squaring but assumed (without checking) that $x = -4$ must be extraneous, since two-radical problems 'usually' produce one bad root, and never actually tested it.

$x = 4$

Simplified $(x + 4) - 4$ incorrectly as $4 - x$ instead of x , giving $(x + 4)(4 - x) = 0$ and roots $x = -4$ or $x = 4$, then reported the unfamiliar-looking root $x = 4$ instead of the correct $x = 0$.

$x = -4$

Stopped after confirming the first valid check on $x = -4$ and never went back to test whether $x = 0$ was also a valid solution worth reporting.

Shortcut: Once you reach $x + 4 = 2\sqrt{x + 4}$, let $u = \sqrt{x + 4}$: the equation becomes $u^2 = 2u$, so $u = 0$ or $u = 2$, and converting back to $x(x = u^2 - 4)$ is one quick line for each.

Desmos Tip: Graph $y = \sqrt{2x + 9} - \sqrt{x + 4}$ and $y = 1$, then find both intersection points — the calculator shows two crossings, at $x = -4$ and $x = 0$, confirming neither is extraneous here.



PART 2

Geometry and Trigonometry

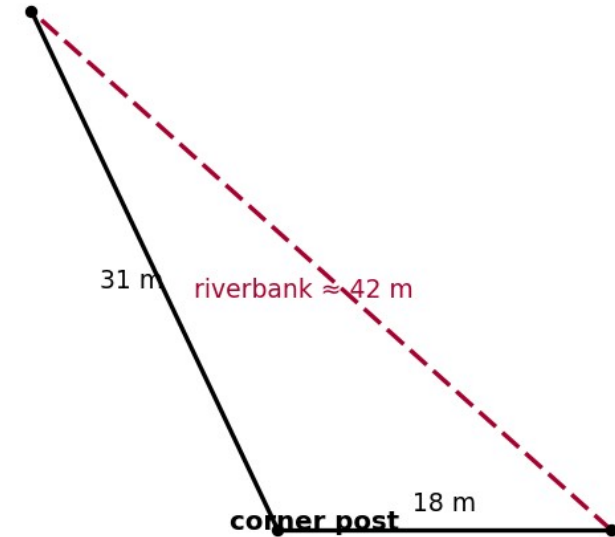
Angles, Parallel Lines & Triangle Fundamentals • Right Triangles & Trigonometry • Radians & the Unit Circle • Circles — Circumference, Arcs, Sectors, Equations & Theorems • Volume, Surface Area & Similarity of Solids

Core Rules — In Detail

- Complementary & Supplementary Angles — Complementary: sum = 90° . Supplementary: sum = 180° . Vertical angles (formed by two intersecting lines) are always equal; adjacent angles share a vertex and a side.
- Parallel Lines Cut by a Transversal — Corresponding, alternate interior, and alternate exterior angle pairs are EQUAL. Same-side interior angle pairs are SUPPLEMENTARY (sum to 180°).
- Triangle Angle Sum — $A + B + C = 180^\circ$ True for every triangle, regardless of shape — knowing any two angles gives you the third instantly.
- Classifying Triangles — By angle: acute (all $<90^\circ$), right (one $=90^\circ$), obtuse (one $>90^\circ$). By side: equilateral (all sides equal), isosceles (two sides equal, base angles equal), scalene (no sides equal).
- Triangle Inequality Theorem — $a + b > c$ (for all three pairs of sides) The sum of any two sides must exceed the third side, or no triangle can be formed — use this to bound a missing side's possible range.
- Side-Angle Relationship — The largest side is always opposite the largest angle, and the smallest side is opposite the smallest angle.
- Exterior Angle Theorem — Exterior angle = sum of the two remote interior angles Formed by extending one side of a triangle; equals the sum of the two angles that are NOT adjacent to it.

Watch Out: Same-side interior angles are the one pair in the parallel-lines diagram that are SUPPLEMENTARY, not equal — mixing this up with corresponding/alternate angles is the most common error. ⚠ The triangle inequality gives you a RANGE ($5 < x < 19$, for example), not a single boundary — check that a candidate side length satisfies BOTH the upper and lower bound.

A landscaping company is fencing a triangular plot of land next to a river in Utrecht. Two sides of the triangle are already fixed: one fence runs 18 meters and the other runs 31 meters, meeting at a shared corner post. The third side runs along the riverbank (unfenced, since the river itself forms the boundary), but for cost-estimation purposes the company treats it as if it were fenced too. The riverside length must be a whole number of meters, and the total perimeter of the plot (all three sides added together) must be a multiple of 7 meters. What is the greatest possible perimeter satisfying all of these conditions?



Formula / Rule Used: Triangle Inequality Theorem: the third side must be strictly between the difference and the sum of the other two sides.

Work through it, then advance for the full step-by-step solution.



Step 1: Let x = the riverside length. By the Triangle Inequality Theorem, x must satisfy $31 - 18 < x < 31 + 18$.

Step 2: Simplify: $13 < x < 49$.

Step 3: The perimeter is $P = 18 + 31 + x = 49 + x$, so shift the inequality by adding 49 to all three parts: $13 + 49 < P < 49 + 49$.

Step 4: This gives $62 < P < 98$.

Step 5: List the multiples of 7 strictly between 62 and 98: 63, 70, 77, 84, 91.

Step 6: The greatest of these is 91.

Step 7: Check: this corresponds to $x = 91 - 49 = 42$, and $13 < 42 < 49$ is true, so a valid triangle exists.

Answer: 91 meters (with riverside length 42 meters)

Tip: For a triangle-inequality range problem, find the valid range for the unknown side first, then shift that same range by the known sides' sum to get the perimeter's range directly — don't recompute the perimeter range from scratch.

Trap: The inequality $13 < x < 49$ is strict (not 'less than or equal to'), so a value at either boundary produces a degenerate triangle (a straight line, not a real triangle) — using $x = 49$ or $x = 13$ anyway is the specific error this problem punishes.

Tempting Wrong Answers — and Exactly What Went Wrong

98 meters

Used $x = 49$, the excluded upper boundary of the range, forgetting the inequality is strict, which actually produces a degenerate (flat) triangle, not a valid one.

63 meters

Found the correct range $62 < P < 98$ but picked the SMALLEST valid multiple of 7 in that range instead of the greatest, misreading 'greatest possible perimeter' as 'least'.

90 meters

Correctly narrowed down to multiples of 7 below 98 but mistakenly selected 90 as the answer, which isn't actually a multiple of 7 ($90/7$ is not a whole number).

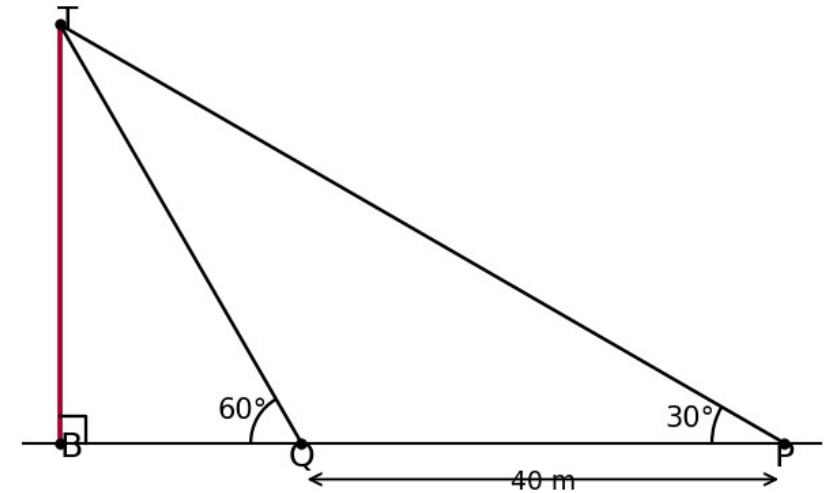
Shortcut: Shift the side-length inequality ($13 < x < 49$) by adding 49 to every part to get the perimeter's range directly ($62 < P < 98$), skipping a separate perimeter recalculation.

Core Rules — In Detail

- The Pythagorean Theorem — $a^2 + b^2 = c^2$ c is always the hypotenuse — the side opposite the right angle, and the longest side of the triangle.
- Pythagorean Triples — 3-4-5, 5-12-13, 8-15-17, 7-24-25 (and any whole-number multiple) Recognizing a scaled triple (e.g. 15-36-39 = $3 \times (5-12-13)$) instantly gives you the third side without computing any squares.
- 45-45-90 Triangle — Legs: x, x . Hypotenuse: $x\sqrt{2}$ An isosceles right triangle — both legs equal, and the hypotenuse is always $\sqrt{2}$ times a leg.
- 30-60-90 Triangle — Sides: $x, x\sqrt{3}, 2x$ (opposite $30^\circ, 60^\circ, 90^\circ$ respectively) The side opposite 30° is the shortest; the hypotenuse (opposite 90°) is always twice that shortest side.
- Similar Triangles — AA, SSS, or SAS similarity Similar triangles have equal corresponding angles and proportional corresponding sides — set up a proportion between matching sides to solve for an unknown.
- Congruent Triangles — SSS, SAS, ASA, AAS, or HL (right triangles only) Congruent triangles are identical in every corresponding side and angle.
- SOH-CAH-TOA — $\sin\theta = \text{Opp}/\text{Hyp}$, $\cos\theta = \text{Adj}/\text{Hyp}$, $\tan\theta = \text{Opp}/\text{Adj}$ Identify which given/unknown side is opposite vs. adjacent to the angle in question before choosing a ratio.
- Complementary Angle & Pythagorean Identities — $\sin(90^\circ - \theta) = \cos\theta$. $\sin^2\theta + \cos^2\theta = 1$. The two acute angles of any right triangle are complementary, which makes sine and cosine of complementary angles equal.

Watch Out: Trigonometric ratios depend ONLY on the angle, never on the triangle's size — this is exactly why they're preserved across similar triangles. ⚠ In a 30-60-90 triangle, the side opposite 60° is $x\sqrt{3}$, NOT $2x$ — confusing which side pairs with which angle is the most common special-triangle error.

A surveyor in the town of Meteora stands at point P on flat ground and measures the angle of elevation to the top of a rock pillar, point T, as 30° . She then walks 40 meters directly toward the base of the pillar, point B (with P, her new position Q, and B all on the same straight line), and measures the new angle of elevation to T from Q as 60° . Points P, Q, and B are all at the same ground elevation, and angle TBP is a right angle. What is the height of the pillar, BT, in meters?



Formula / Rule Used: 30-60-90 triangle side ratios combined with setting up two right triangles that share the same height.

Work through it, then advance for the full step-by-step solution.

Step 1: Let $h = BT$ (the height of the pillar) and let $x = QB$ (the horizontal distance from the second position to the base).

Step 2: In right triangle TBQ, angle TQB = 60° , so this is a 30-60-90 triangle with the side opposite 60° (which is h) related to the side opposite 30° ... instead use tangent directly: $\tan(60^\circ) = h/x$, so $h = x \cdot \sqrt{3}$.

Step 3: In right triangle TBP, angle TPB = 30° , and $PB = PQ + QB = 40 + x$.

Step 4: Apply tangent again: $\tan(30^\circ) = h/(40 + x)$, so $h = (40 + x) \cdot (1/\sqrt{3})$.

Step 5: Set the two expressions for h equal: $x \cdot \sqrt{3} = (40 + x)/\sqrt{3}$.

Step 6: Multiply both sides by $\sqrt{3}$: $3x = 40 + x$.

Step 7: Solve for x : $2x = 40$, so $x = 20$.

Step 8: Substitute back into $h = x \cdot \sqrt{3}$: $h = 20 \cdot \sqrt{3}$ meters (approximately 34.6 meters).

Answer: $BT = 20 \cdot \sqrt{3}$ meters (approximately 34.6 m)

Tip: For two-station angle-of-elevation problems, define the unknown height and the unknown near-distance as two variables, write tangent equations for both triangles, and solve the resulting system — don't try to jump straight to the answer with one triangle.

Trap: A student often tries to use only the far triangle (angle 30° , distance 40) and solves $h = 40 \cdot \tan(30^\circ) = 40/\sqrt{3}$, ignoring that 40 is NOT the full distance from P to B (it's only the distance she walked, not PB), producing a wrong height.

Tempting Wrong Answers — and Exactly What Went Wrong

BT = $40/\sqrt{3}$ meters (approx 23.1 m)

Incorrectly treated the 40-meter walk as the full distance PB from the original point to the base, using $\tan(30^\circ) = h/40$ instead of setting up the two-triangle system.

BT = $40\sqrt{3}$ meters (approx 69.3 m)

Incorrectly treated the 40-meter walk as QB (the distance from the second position to the base) and used $\tan(60^\circ) = h/40$ directly, skipping the system entirely.

BT = 20 meters

Solved the system correctly to find $x = 20$ but then reported x itself as the final answer instead of substituting back to find $h = x\sqrt{3}$.

Shortcut: Once you notice the two angles are 30° and 60° (complementary-looking but really just both present), the ratio $\tan(60^\circ)/\tan(30^\circ) = 3$ tells you PB is exactly 3 times QB, so $x = 40/(3-1) = 20$ in one line without full algebra.

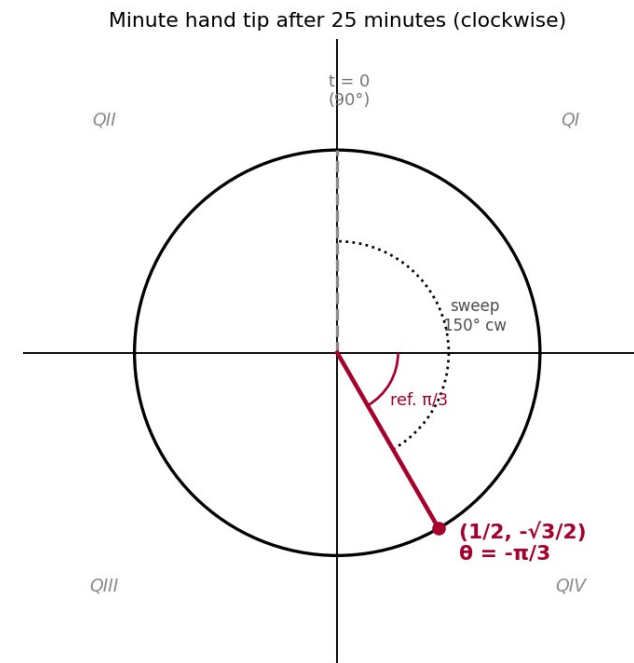
Desmos Tip: Graph the two lines $y = x \cdot \tan(60^\circ)$ and $y = (x+40) \cdot \tan(30^\circ)$ using x as the horizontal distance from Q, and find their intersection point's y -value to confirm $h = 20\sqrt{3}$.

Core Rules — In Detail

- Degree-Radian Conversion — π radians = 180° Multiply degrees by $\pi/180$ to get radians; multiply radians by $180/\pi$ to get degrees.
- Common Angles in Radians — $30^\circ=\pi/6$, $45^\circ=\pi/4$, $60^\circ=\pi/3$, $90^\circ=\pi/2$, $180^\circ=\pi$, $360^\circ=2\pi$ Memorizing this table lets you convert instantly without setting up a proportion each time.
- The Unit Circle — Point at angle $\theta = (\cos\theta, \sin\theta)$ A circle of radius 1 centered at the origin — the x-coordinate of the point at angle θ is always $\cos\theta$, and the y-coordinate is always $\sin\theta$.
- Quadrant Signs — QI: (+,+) QII: (-,+) QIII: (-,-) QIV: (+,-) Use the reference angle (the acute angle to the x-axis) to find the magnitude, then apply the correct sign for the quadrant.
- Coterminal Angles — Add or subtract 360° (or 2π) repeatedly Coterminal angles point in the same direction and share the same trig values; use this to bring any angle into the standard $[0^\circ, 360^\circ)$ or $[0, 2\pi)$ range.

Watch Out: A negative angle or an angle over 360° is not 'wrong' — just add or subtract full rotations (360° or 2π) until it lands in the standard range, then read off the trig values normally. ⚠ The sign of cosine and sine depends entirely on the quadrant — always determine the quadrant before assigning + or - to the reference-angle value.

A large decorative clock in a plaza in Vienna has a minute hand modeled as a point moving on a unit circle (radius 1) centered at the clock's center. At time $t = 0$ minutes, the tip of the minute hand is at the top of the clock, corresponding to standard position angle $\pi/2$. The minute hand moves clockwise at a constant rate, completing one full revolution (2π radians) every 60 minutes. What are the exact coordinates of the tip of the minute hand at $t = 25$ minutes?



Formula / Rule Used: Angular rate = 2π / period; clockwise rotation subtracts angle from the starting position.

Work through it, then advance for the full step-by-step solution.

Step 1: Find the rotation rate: one full revolution (2π radians) every 60 minutes gives $2\pi/60 = \pi/30$ radians per minute.

Step 2: Find the angle swept in 25 minutes: $25 \times \pi/30 = 25\pi/30 = 5\pi/6$ radians.

Step 3: Since the hand moves clockwise, subtract this swept angle from the starting angle instead of adding it: new angle = $\pi/2 - 5\pi/6$.

Step 4: Use a common denominator of 6: $\pi/2 = 3\pi/6$, so new angle = $3\pi/6 - 5\pi/6 = -2\pi/6 = -\pi/3$.

Step 5: The angle $-\pi/3$ has reference angle $\pi/3$ and lies in Quadrant IV (right of the y-axis, below the x-axis).

Step 6: In Quadrant IV, cosine is positive and sine is negative: $\cos(-\pi/3) = \cos(\pi/3) = \frac{1}{2}$, and $\sin(-\pi/3) = -\sin(\pi/3) = -\sqrt{3}/2$.

Step 7: So the tip of the minute hand is at $(\frac{1}{2}, -\sqrt{3}/2)$.

Answer: $(\frac{1}{2}, -\sqrt{3}/2)$

Tip: When a rotation is described as clockwise, subtract the swept angle from the starting angle instead of adding — direction sets the sign before you touch magnitude.

Trap: A rushed student adds $(5\pi)/6$ to $\pi/2$ instead of subtracting (forgetting that 'clockwise' reverses the usual counterclockwise-positive convention), landing at $(4\pi)/3$ and getting $(-\frac{1}{2}, -\sqrt{3}/2)$ instead of the correct $(\frac{1}{2}, -\sqrt{3}/2)$.

Tempting Wrong Answers — and Exactly What Went Wrong

$(-\frac{1}{2}), -\sqrt{\frac{3}{2}}$

Added the swept angle $5\pi/6$ to the start $\pi/2$ instead of subtracting for clockwise motion, landing at $4\pi/3$ instead of $-\pi/3$.

$(-\frac{1}{2}), \sqrt{\frac{3}{2}}$

Correctly subtracted to get $-\pi/3$ but then applied Quadrant II sign rules (cos negative, sin positive) instead of Quadrant IV, flipping both signs.

$(\frac{1}{2}), \sqrt{\frac{3}{2}}$

Found the correct reference angle $\pi/3$ but forgot the hand is below the x-axis at $t = 25$ min, so left sine positive instead of negative.

Shortcut: Since $25/60 = 5/12$ of a full revolution, jump straight to $5/12 \times 360^\circ = 150^\circ$ clockwise from 90° , i.e. an angle of -60° , without ever computing the $\pi/30$ -per-minute rate explicitly.

Desmos Tip: Type the point $(\cos(-\pi/3), \sin(-\pi/3))$ into the calculator in radian mode to confirm it evaluates to $(0.5, -0.866\dots)$.

Core Rules — In Detail

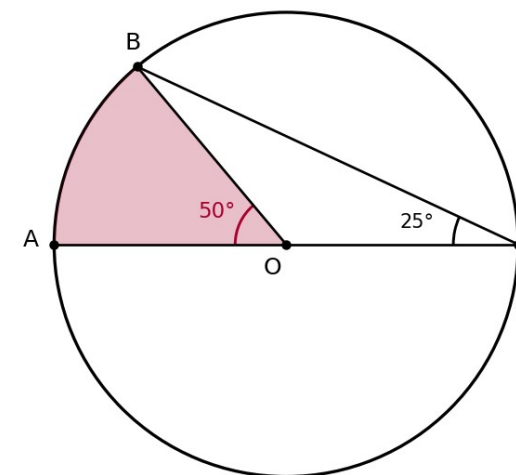
- Circumference and Area — $C = 2\pi r = \pi d$. $A = \pi r^2$ Diameter is twice the radius; both formulas depend only on r (or d).
- Arc Length — Arc length = $(\theta/360^\circ) \times 2\pi r$ An arc is exactly that same fraction of the full circumference that its central angle θ is of 360° .
- Sector Area — Sector area = $(\theta/360^\circ) \times \pi r^2$ Same proportional logic as arc length, applied to the full circle's area instead of its circumference.
- Equation of a Circle — $(x - h)^2 + (y - k)^2 = r^2$ Center is at (h, k) — note both signs are subtracted even though the center coordinates might be negative (e.g. center $(-3, 2)$ still uses $(x - (-3))^2 = (x + 3)^2$).
- Completing the Square for a Circle — Group x -terms and y -terms separately, complete the square for each (adding the same amount to both sides), then read off the center and radius from the resulting standard form.
- Tangent Lines — Radius $\perp$ tangent line A tangent touches the circle at exactly one point, and the radius drawn to that point is always perpendicular to the tangent — this creates a right triangle you can solve with the Pythagorean theorem.
- Inscribed Angle Theorem — Inscribed angle = $(1/2) \times$ intercepted arc An angle inscribed in a semicircle (intercepting a diameter) is always exactly 90° ; two inscribed angles that intercept the same arc are equal to each other.

Watch Out: In the circle equation, the center's y -coordinate sign gets flipped inside the formula: a center at (h, k) still appears as $(x - h)^2 + (y - k)^2$ even when h or k is negative. ⚠ Arc length and sector area both use $\theta/360^\circ$ as the fraction — using the arc's chord length or any other measurement instead of the central angle is a common setup error.

Worked Problem · Very Hard

Circle O has a radius of 18 meters, and segment AC is a diameter of circle O. Points A, B, and C all lie on circle O, and the inscribed angle ACB (with vertex C, intercepting the minor arc AB that does not contain C) measures 25° . What is the length of minor arc AB, in meters, in terms of π ?

Circle O, $r = 18$ m — inscribed $\angle ACB = 25^\circ$



arc AB = 5π meters

Formula / Rule Used: Inscribed Angle Theorem combined with the Arc Length formula.

Work through it, then advance for the full step-by-step solution.

Step 1: By the Inscribed Angle Theorem, an inscribed angle is half the measure of the arc it intercepts, so the central angle AOB is double the inscribed angle ACB.

Step 2: Central angle AOB = $2(25^\circ) = 50^\circ$.

Step 3: (Note: AC being a diameter is extra information here — it is not needed to find arc AB, since angle ACB already gives the intercepted arc directly.)

Step 4: Apply the arc length formula: arc length = $(\theta/360^\circ) \times 2\pi r$.

Step 5: Substitute $\theta = 50^\circ$ and $r = 18$: arc length = $(50/360) \times 2\pi(18) = (50/360) \times 36\pi$.

Step 6: Simplify: $(50 \times 36)/360 = 1800/360 = 5$.

Step 7: Arc length AB = 5π meters.

Answer: Arc AB = 5π meters

Tip: Whenever an inscribed angle is given, always double it first to get the central angle before touching the arc length formula — the inscribed angle itself is never the θ you plug in.

Trap: A rushed student plugs the inscribed angle (25°) directly into the arc length formula as if it were the central angle, skipping the required doubling step from the Inscribed Angle Theorem.



Tempting Wrong Answers — and Exactly What Went Wrong

2.5π meters

Used the inscribed angle (25°) directly as the central angle instead of doubling it per the Inscribed Angle Theorem, computing $(25/360) \times 2\pi(18) = 2.5\pi$ instead of 5π .

10π meters

Mistook the given radius (18) for a diameter and doubled it to 36 before applying the arc length formula, instead of using 18 as the radius directly.

50

Correctly doubled the inscribed angle to find the central angle (50°) but stopped there, reporting the angle measure itself instead of converting it into an arc length.

Shortcut: Combine the doubling and the arc length fraction into one line — $\text{arc} = (2 \times 25 / 360) \times 2\pi(18)$ — instead of writing the central angle as a separate step.

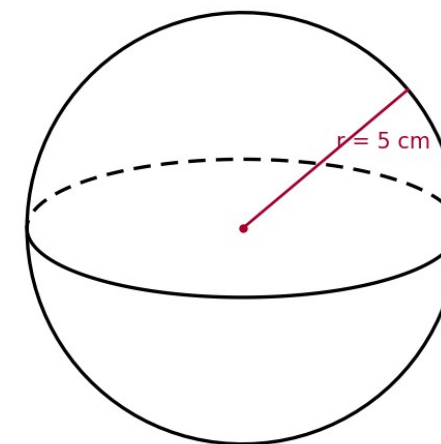
Core Rules — In Detail

- Prisms and Cylinders — $V = (\text{area of base}) \times \text{height}$. Rectangular prism: $V=lwh$. Cylinder: $V=\pi r^2 h$ Any solid with two parallel, congruent bases and straight sides uses this same 'base times height' logic.
- Pyramids and Cones — $V = (1/3) \times (\text{area of base}) \times \text{height}$. Cone: $V=(1/3)\pi r^2 h$. Square pyramid: $V=(1/3)s^2 h$ A pyramid or cone always holds exactly 1/3 the volume of a prism/cylinder with the same base and height.
- Spheres — $V = (4/3)\pi r^3$. Surface area = $4\pi r^2$ Both formulas depend only on the radius.
- Surface Area of Common Solids — Cube: $S=6s^2$. Rect. prism: $S=2(ab+bc+ac)$. Cylinder: $S=2\pi rh+2\pi r^2$ Surface area adds up the area of every face (or the lateral surface plus the two circular bases, for a cylinder).
- Composite Solids — Break the solid into recognizable pieces (e.g. a cylinder plus a hemisphere), find each volume separately, and add (or subtract, for a hollowed-out shape).
- Polygon Interior Angle Sum — $(n - 2) \times 180^\circ$ n is the number of sides; a triangle ($n=3$) gives 180° , a quadrilateral ($n=4$) gives 360° , and so on.
- Similarity Ratios — Perimeter ratio = λ . Area ratio = λ^2 . Volume ratio = λ^3 If two shapes are similar with scale factor λ , every LENGTH scales by λ , every AREA scales by λ^2 , and every VOLUME scales by λ^3 — this lets you jump between volume, surface area, and length ratios without finding actual dimensions.

Watch Out: Squaring or cubing the scale factor (not the raw dimension) is the key move for similarity problems — a volume ratio of 8 means a LENGTH ratio of 2 (since $2^3=8$), not a length ratio of 8. ⚠ A hemisphere's volume is exactly HALF a full sphere's — remember to divide $(4/3)\pi r^3$ by 2, not just reuse the sphere formula directly, in a composite-solid problem.

A spherical marker buoy manufactured by a shipping company in Bergen has a total surface area of 314 square centimeters. Using $\pi \approx 3.14$, what is the volume of the buoy, in cubic centimeters, rounded to the nearest whole number?

Spherical Buoy



$$\text{Surface Area} = 4\pi r^2 = 314 \text{ cm}^2$$

Formula / Rule Used: Sphere surface area $SA = 4\pi r^2$ solved for r , then substituted into sphere volume $V = (\frac{4}{3})\pi r^3$.

Work through it, then advance for the full step-by-step solution.

Step 1: Set up the surface area equation: $4\pi r^2 = 314$.

Step 2: Substitute $\pi \approx 3.14$: $4(3.14)r^2 = 314$, so $12.56r^2 = 314$.

Step 3: Solve for r^2 : $r^2 = \frac{314}{12.56} = 25$.

Step 4: Take the square root to find r : $r = \sqrt{25} = 5$ centimeters.

Step 5: Substitute $r = 5$ into the volume formula: $V = \left(\frac{4}{3}\right)\pi r^3 = \left(\frac{4}{3}\right)(3.14)(5^3) = \left(\frac{4}{3}\right)(3.14)(125)$.

Step 6: Compute: $\left(\frac{4}{3}\right)(3.14)(125) = \left(\frac{4}{3}\right)(392.5) = 523.33$, which rounds to 523 cubic centimeters.

Answer: 523 cubic centimeters

Tip: When surface area is given but volume is asked for, always solve for the radius first (undoing the square), then plug that radius into the volume formula — never skip straight from the SA equation to a volume calculation.

Trap: A student who solves $r^2 = 25$ but forgets to take the square root often plugs $r^2 = 25$ directly into the volume formula as if it were r itself, producing a wildly inflated answer.

Tempting Wrong Answers — and Exactly What Went Wrong

65,417 cubic centimeters

Solved $r^2 = 25$ correctly but skipped the square root step, substituting 25 in place of r into $V = \frac{4}{3}\pi r^3$.

4,187 cubic centimeters

Found $r = 5$ correctly but then doubled it to a diameter of 10 before plugging into the volume formula, confusing radius with diameter.

1,570 cubic centimeters

Found $r = 5$ correctly but used $V = 4\pi r^3$ (the surface-area coefficient of 4) instead of $V = \frac{4}{3}\pi r^3$, dropping the $\frac{1}{3}$ factor.

Shortcut: Once r is known, note that $V = \frac{r}{3} \times SA$, since $\frac{4}{3}\pi r^3 = \frac{r}{3}(4\pi r^2)$; plugging in gives $V = \frac{5}{3}(314) \approx 523.3$ without recomputing r^3 from scratch.

Desmos Tip: Use the calculator's solver to find r from $4(3.14)r^2 = 314$, then compute $\frac{4}{3}(3.14)r^3$ with that r in a second line.